

Siddhika Sriram

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EDUCATION

Masters in Computer Science

Pennsylvania State University

Harrisburg, PA

Coursework: Database Management Systems | Advanced Operating Systems, Artificial Intelligence, Machine Learning, Natural Language processing | Data mining | Design and analysis of algorithms

Bachelors in Information Technology

Anna University

Coimbatore, India

SKILLS

Programming Languages: Python, SQL, R, C, C++

Practices: Generative AI (GenAI), ETL, EDA, Agile, CI/CD

Data Platforms & Tools: Databricks, PySpark, Airflow, Docker, Kubernetes

Visualization Tools: Tableau, Power BI

Libraries: TensorFlow, Keras, SciKit-Learn, Pandas, NumPy, Matplotlib, Plotly

Statistics & Analysis: A/B Testing, Hypothesis Testing, Statistical Data Analysis

Certifications: AWS Cloud Practitioner (In Progress)

WORK EXPERIENCE

Data Engineer (Contract)

11/2024 - Present

Guardian Life

New York, NY

- Built and optimized data pipelines using Python, Pandas, and NumPy to preprocess large-scale claim, medical, and occupational datasets for XGBoost-based claim closure prediction models.
- Integrated prediction outputs with SQL databases and automated report generation in CSV and Excel formats, enabling efficient delivery of actionable insights.
- Collaborated with cross-functional teams to enhance data workflows, ensuring accurate feature engineering and seamless integration with business processes.

Data Scientist (Full Time)

01/2020 - 05/2022

Bosch

Coimbatore, India

- Developed an LSTM-based stock price prediction model using time series forecasting, achieving 90% accuracy.
- Integrated API to dynamically fetch and present relevant news to offer context to stock price trends.
- Built an employee resource management dashboard in Tableau, enhancing resource utilization thereby improving project-employee competency mapping by 15% through exploratory data analysis (EDA).
- Designed and maintained SQL queries and data pipelines for real-time data extraction and transformation, enabling faster decision-making.
- Supported early-stage startups by delivering data-driven market analyses, aiding healthcare product development.

RESEARCH EXPERIENCE

Research Assistant

08/2023 - 08/2024

Penn State University

Harrisburg, PA

- Designed a chatbot using Retrieval-Augmented Generation (RAG) for unstructured and semi-structured enterprise data across SQL and NoSQL databases.
- Engineered data pipelines and developed word embeddings stored in a Chroma vector database, enabling real-time cross-application data retrieval.
- Conducted market basket analysis on 10,000+ grocery transactions using the Apriori algorithm, resulting in optimized product placements.

ML Researcher

06/2021 - 12/2023

IUCAA

Pune, India

- Developed AI systems for analyzing 50,000+ LASER mode images from a gravitational wave interferometer, improving classification precision by 30%.
- Built a custom CNN model for astronomical data classification, achieving 92% accuracy through Bayesian inference and deep learning techniques.
- Enhanced model robustness with data augmentation, leading to improved classification model performance.

PROJECTS

College Cafeteria Sales Enhancement | Python, SQL, PowerBI

- Analyzed cafeteria sales data using SQL and performed EDA to identify trends, resulting in a 5% increase in sales. Automated data ingestion and transformation pipelines resulting in a 45% reduction in manual data cleaning and feature engineering efforts.

Impulse Product Analysis and Demand Forecasting | Python, SQL, Tableau

- Conducted customer interaction analysis on impulse products. Forecasted demand for impulse products using customer interaction data, reducing inventory discrepancies by 25%.

Brain Tumor Segmentation using Federated Learning | Python, AWS, Tensorflow

- Developed a brain tumor segmentation model using Federated Learning, ensuring privacy by training on distributed client data. Implemented the FedAvg Optimization algorithm for iterative accuracy improvements, aggregating model weights centrally on AWS.